



# NORTH SYDNEY BOYS HIGH SCHOOL

## EXTENSION 2 MATHEMATICS

HSC Course Assessment Task 1

December, 2023

### General instructions

- Working time – 60 minutes.  
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.

### QUESTIONS 1-4

- Commence each new question on a NEW BOOKLET.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: ..... # BOOKLETS USED: .....

Teacher (please ✓)

- ☐ Mr Berry
- ☐ Ms Lee
- ☐ Mr Umakanthan
- ☐ Dr Vranesevic

Marker's use only.

| QUESTION | 1               | 2               | 3               | 4               | Total           |
|----------|-----------------|-----------------|-----------------|-----------------|-----------------|
| MARKS    | $\overline{12}$ | $\overline{10}$ | $\overline{12}$ | $\overline{13}$ | $\overline{47}$ |

| Question 1 (12 Marks) |                                                                                                                                          | Marks |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------|
| a)                    | If $(1 + i)z = 3 - i$ , find:                                                                                                            |       |
| i)                    | $z$ in the form $x + iy$                                                                                                                 | 2     |
| ii)                   | $ z $                                                                                                                                    | 1     |
| iii)                  | $\text{Arg } z$                                                                                                                          | 1     |
| iv)                   | $\bar{z}$                                                                                                                                | 1     |
| b)                    |                                                                                                                                          |       |
| i)                    | Find $\sqrt{8 + 6i}$                                                                                                                     | 2     |
| ii)                   | Solve $z^2 + (2 + 4i) - 11 - 2i = 0$                                                                                                     | 2     |
| c)                    | $z$ is a complex number such that $z = k(\cos \theta + i \sin \theta)$ where $k \in \mathbb{R}$<br>Show $\arg(z + k) = \frac{\theta}{2}$ | 3     |

| Question 2 (10 Marks) |                                                                                                       | Marks |
|-----------------------|-------------------------------------------------------------------------------------------------------|-------|
| a)                    | Consider the statement “for all integers $a$ and $b$ , if $a + b$ is even, then $a$ and $b$ are even” |       |
| i)                    | Write the contrapositive of the statement                                                             | 1     |
| ii)                   | Write the converse of the statement                                                                   | 1     |
| iii)                  | Write the negation of the statement                                                                   | 1     |
| iv)                   | Is the original statement true or false? Prove the statement is true/false                            | 1     |
| b)                    | State the negation of<br>$(\forall x \in \mathbb{Z})(x = 3 \Rightarrow x^2 = 9)$                      | 1     |
| c)                    | Prove that if $n > 0$ and $4^n - 1$ is prime, then $n$ is odd                                         | 3     |
| d)                    | Prove that there are no integers $x$ and $y$ such that $x^2 = 4y + 2$                                 | 2     |

### Question 3 (12 Marks)

Marks

a)

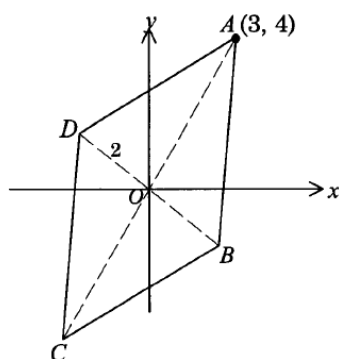


Diagram is not to scale

In the diagram,  $ABCD$  is a rhombus whose diagonals meet at  $O$ , the origin.  $A$  represents the complex number  $3 + 4i$  and  $OD = 2$  units.

Find the complex number represented by

- |     |     |   |
|-----|-----|---|
| i)  | $C$ | 1 |
| ii) | $D$ | 2 |

b) The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

Where  $p$  and  $q$  are real, has roots  $\alpha, \beta$  and  $\gamma$ .

It is given that  $\alpha = 2 + 2\sqrt{3}i$

- |      |                                                            |   |
|------|------------------------------------------------------------|---|
| i)   | Write down another root, $\beta$ , of the equation         | 1 |
| ii)  | Find the third root, $\gamma$                              | 2 |
| iii) | Find the values of $p$ and $q$                             | 2 |
| iv)  | By expressing $\alpha$ in modulus-argument form, show that | 2 |

$$(2 + 2\sqrt{3}i)^n = 4^n \left( \cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

- |    |                  |   |
|----|------------------|---|
| v) | Hence, show that | 2 |
|----|------------------|---|

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{n\pi}{3} + \left( -\frac{1}{2} \right)^n$$

where  $n$  is an integer

### Question 4 (13 Marks)

Marks

- |     |                                                                                                                                                                 |   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| a)  | Find the five roots of unity, and then express the complex roots in terms of $\omega$ , where $\omega$ is the complex root with the smallest positive argument. | 2 |
| b)  | Prove that $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) = 5$                                                                                         | 2 |
| c)  | Show that $(1 - \omega)(1 - \omega^4) = 2 - 2 \cos \frac{2\pi}{5}$                                                                                              | 2 |
| d)  | Hence or otherwise, show that $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} = \frac{\sqrt{5}}{4}$                                                                     | 3 |
| e)  | Suppose $P_0, P_1, P_2, P_3, P_4$ are the corresponding points of $1, \omega, \omega^2, \omega^3$ and $\omega^4$ in the Argand plane                            |   |
| i)  | Show that $ \overrightarrow{P_0P_1}  = 2 \sin \frac{\pi}{5}$                                                                                                    | 2 |
| ii) | Hence find the value of                                                                                                                                         | 2 |

$$|\overrightarrow{P_0P_1}| \times |\overrightarrow{P_0P_2}| \times |\overrightarrow{P_0P_3}| \times |\overrightarrow{P_0P_4}|$$

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Q1

a) If  $(1+i)z = 3-i$  find  $z$ , alternately;

i)  $z$  in the form  $x+iy$

$$z = \frac{3-i}{1+i} \times \frac{1-i}{1-i} \quad \checkmark \quad (1+i)z = 3-i$$

$$= \frac{3-i-3i-1}{1+1}$$

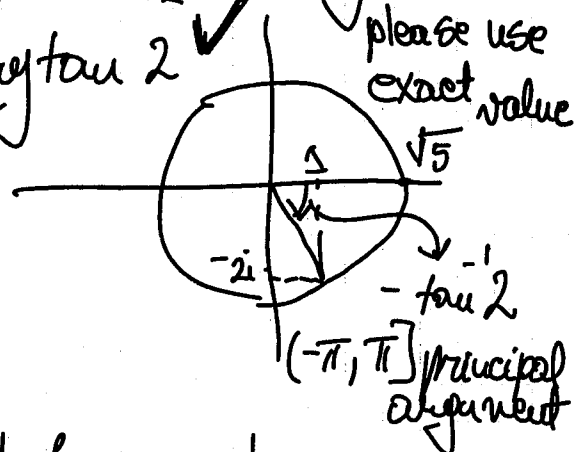
$$= \frac{2-4i}{2}$$

$$= 1-2i \quad \checkmark$$

ii)  $|z| = \sqrt{1^2 + 2^2}$   
 $= \sqrt{5} \quad \checkmark$

iii)  $\text{Arg } z = \arg \tan \frac{-2}{1} = \arg \tan(-2)$   
 $= -\arg \tan 2 \quad \checkmark$

iv)  $\bar{z} = \overline{1-2i}$   
 $= 1+2i \quad \checkmark$



CFE - carried forward error considered

b) Find  $\sqrt{8+6i}$

$$\sqrt{8+6i} = x+iy$$

$$8+6i = x^2-y^2 + 2xyi$$

①  $x^2-y^2=8$

②  $2xy=6 \Rightarrow x = \frac{3}{y}$

②  $\Rightarrow$  ①  $\Rightarrow \frac{9}{y^2} - y^2 = 8$

$$9 - y^4 - 8y^2 = 0$$

$$y^4 + 8y^2 - 9 = 0$$

$$(y^2-1)(y^2+9) = 0$$

$$y = \pm 1, y = \pm 3i \notin \mathbb{R}$$

$$\therefore y = \pm 1 \Rightarrow x = \pm 3$$

$$\therefore \pm(3+i) \quad \checkmark$$

Alternately:

①  $x^2-y^2=8$

②  $x^2+y^2 = \sqrt{8^2+6^2} = 10$

①+②:

$$2x^2 = 18$$

$$x = \pm 3$$

②-①

$$2y^2 = 2$$

$$y = \pm 1$$

$$\therefore \sqrt{8+6i} = \pm(3+i)$$

ii) Solve  $z^2 + (2+4i)z - 11 - 2i = 0$

$$\Delta = b^2 - 4ac$$

$$= (2+4i)^2 + 4 \times 1 \times (11+2i)$$

$$= 4 + 16i - 16 + 44 + 8i$$

$$= 32 + 24i$$

$$= 4(8+6i)$$

$$z = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$= \frac{-(2+4i) \pm \sqrt{4(8+6i)}}{2 \times 1}$$

$$= \frac{-2-4i \pm 2(3+i)}{2} \quad \text{from (i)}$$

$$= -1-2i \pm (3+i)$$

$$= 2-i \quad \checkmark \quad \text{or} \quad -4-3i \quad \checkmark$$

c)  $z$  is complex number such as  
 $z = k(\cos \theta + i \sin \theta)$ ,  $k \in \mathbb{R}$ . Show  $\arg(z+k) = \frac{\theta}{2}$

Solution 1:

$$z = k(\cos \theta + i \sin \theta), k \in \mathbb{R}$$

$$z+k = k(\cos \theta + i \sin \theta) + k$$

$$= k(\cos \theta + 1) + i k \sin \theta$$

$$\arg(z+k) = \tan^{-1} \left( \frac{k \sin \theta}{k(\cos \theta + 1)} \right)$$

$$= \tan^{-1} \left( \frac{\sin \theta}{1 + \cos \theta} \right)$$

$$= \tan^{-1} \left( \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2} - 1 + 1} \right)$$

$$= \tan^{-1} \left( \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right)$$

$$= \tan^{-1} \left( \tan \frac{\theta}{2} \right)$$

$$= \frac{\theta}{2}$$

$$\begin{aligned} \sin \theta &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ \cos \theta &= \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \\ &= \cos^2 \frac{\theta}{2} - 1 + \cos^2 \frac{\theta}{2} \\ &= 2 \cos^2 \frac{\theta}{2} - 1 \end{aligned}$$

Solution 2,

$$\arg(z+k) = \arg(k(\cos\theta + i\sin\theta + 1))$$

$$= \arg(\cos\theta + i\sin\theta + 1)$$

$$= \arg(e^{i\theta} + 1)$$

$$= \arg(e^{i\theta/2} (e^{i\theta/2} + e^{-i\theta/2}))$$

$$= \arg(e^{i\theta/2} \times 2 \cos \frac{\theta}{2})$$

real part

$$= \arg e^{i\theta/2} + \arg(2 \cos \frac{\theta}{2})$$

$$= \frac{\theta}{2}$$

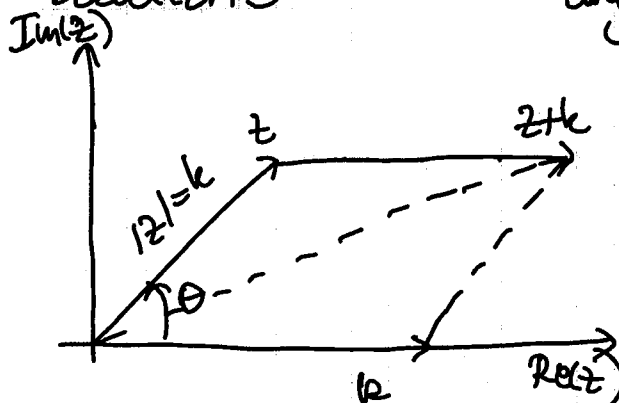
as for  $z = e^{i\theta} \Rightarrow z^n = e^{in\theta} = \cos(n\theta) + i\sin(n\theta)$   
 $\bar{z} = e^{-i\theta} \Rightarrow \bar{z}^n = e^{-in\theta} = \cos(n\theta) - i\sin(n\theta)$

$$e^{in\theta} + e^{-in\theta} = 2 \cos(n\theta)$$

in our case  $n = \frac{1}{2}$

$$\therefore e^{i\theta/2} + e^{-i\theta/2} = 2 \cos \frac{\theta}{2}$$

Solution 3



$$\arg z = \theta$$

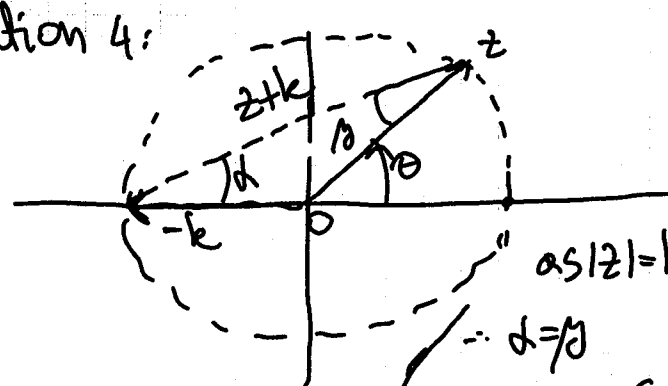
$$|z| = k$$

$$k \in \mathbb{R}$$

$z+k$  form a rhombus

$\therefore$  angle in a rhombus is bisected by its diagonals  $\therefore \arg(z+k) = \frac{\theta}{2}$

Solution 4:



$$|z| = k$$

$$\arg(z) = \theta$$

$$\text{as } |z| = k \text{ \& } |-k| = k$$

$$\therefore \angle = \theta$$

$\angle = \theta$  Exterior angle theorem

$$\therefore 2\theta = \theta$$

$$\therefore \theta = \frac{\theta}{2}, \arg(z+k) = \frac{\theta}{2}$$

or:  
from circle geometry  
angle at the center is  
double the angle  
at circumference with the same arc

## Question 2

a)

i) For all integers  $a$  and  $b$ , if  $a$  or  $b$  is odd, then  $a+b$  is odd

ii) For all integers  $a$  and  $b$ , if  $a$  and  $b$  are even, then  $a+b$  is even

iii) There exists integers  $a$  and  $b$  such that  $a+b$  is even and  $a$  or  $b$  is odd

iv) False. Counterexample:  $a=3$  and  $b=5$

$a+b=8$  but neither  $a$  nor  $b$  are even.

b)  $(\exists x \in \mathbb{Z})(x=3 \wedge x^2 \neq 9)$

c) The contrapositive is proven instead. That is:

If  $n$  is even, then  $4^n - 1$  is not prime.

Let  $n = 2k$ ,  $k \in \mathbb{Z}$ , then  $n$  is even and

$$\begin{aligned} 4^n - 1 &= 4^{2k} - 1 \\ &= (4^k + 1)(4^k - 1) \end{aligned}$$

which is a product of two integers of which neither is 1 ( $n > 0$ ) and is thus not prime.

Hence, by the contrapositive, if  $4^n - 1$  is prime, then  $n$  is odd.

d) By way of contradiction, assume there are integers  $x$  and  $y$

such that  $x^2 = 4y + 2$  (\*)

Now  $4y + 2 = 2(y+1)$  which is even, so  $x^2$  is even.

Let  $x = 2k$ ,  $k \in \mathbb{Z}$

$$\begin{aligned} (2k)^2 &= 4y + 2 \\ 4k^2 &= 4y + 2 \\ 2k^2 &= 2y + 1 \end{aligned}$$

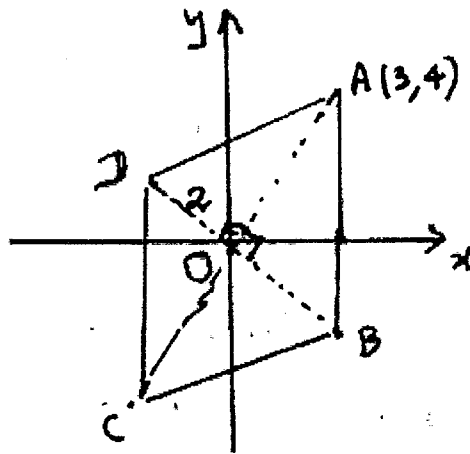
LHS is even and RHS is odd.

Even  $\neq$  Odd

Thus we have a contradiction, so there must not be any integers  $x$  and  $y$  such that  $x^2 = 4y + 2$

Q3

(a)



(i) Diagonals bisect in a Rhombus

$$\begin{array}{l|l} \therefore \vec{OC} = -\vec{OA} & \vec{OC} = \vec{OA} \operatorname{cis} \pi \\ = -(3+4i) & = -1(3+4i) \\ = \underline{\underline{-3-4i}} & = \underline{\underline{-3-4i}} \end{array}$$

Diagonals bisect + correct Answer for  
No reason no mark 1 mark

$$(ii) |\vec{OA}| = 5, |\vec{OC}| = \sqrt{3^2+4^2} = 5$$

$$\Rightarrow \vec{OC} = \frac{2}{5} \vec{OA} \operatorname{cis} \frac{\pi}{2} \quad \checkmark \quad (\text{diagonals } \perp \text{ in a rhombus})$$

$$= \frac{2}{5} (3+4i)i$$

$$= \underline{\underline{-\frac{8}{5} + \frac{6i}{5}}} \quad \checkmark$$

(b)  $2z^3 + pz^2 + qz + 16 = 0 \quad p, q \in \mathbb{R}$   
 roots  $\alpha, \beta$  and  $\gamma$

(i)  $\alpha = 2 + 2\sqrt{3}i$

Since the coefficients are all real,  
 the complex conjugate of  $\alpha$  is also a root.

let  $\beta = \bar{\alpha} = \underline{2 - 2\sqrt{3}i}$  ✓ (correct answer)  
 (Reason: given the benefit of doubt)

(ii) Product of roots  $\alpha\beta\gamma = -\frac{16}{2} = -8$

$\Rightarrow \alpha\bar{\alpha}\gamma = -8$  ✓

$\Rightarrow [2^2 + (2\sqrt{3})^2]\gamma = -8$

$\Rightarrow \gamma = -\frac{8}{16} = -\frac{1}{2}$  ✓

(iii) Sum of roots  $\alpha + \beta + \gamma = -\frac{p}{2}$

$\alpha + \bar{\alpha} + \gamma = -\frac{p}{2}$

$2(2) + (-\frac{1}{2}) = -\frac{p}{2}$

$\Rightarrow \underline{p = -7}$  ✓

~~sum~~ sum of product of 2 roots at a time =  $\frac{q}{2}$

$\Rightarrow \alpha\bar{\alpha} + \bar{\alpha}\gamma + \gamma\alpha = \frac{q}{2}$

$16 + (4)(-\frac{1}{2}) = \frac{q}{2}$

$\Rightarrow \underline{q = 28}$  ✓

← (or by factorising)

$$\begin{aligned}
 \text{(iv)} \quad \alpha &= 2 + 2\sqrt{3}i \\
 &= 4\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\
 &= 4 \operatorname{cis}\left(\frac{\pi}{3}\right) \checkmark \\
 \alpha^n &= 4^n \operatorname{cis}\left(\frac{n\pi}{3}\right) \checkmark \text{ (de Moivre's theorem.)} \\
 \therefore \beta^n &= (\bar{\alpha})^n = 4^n \operatorname{cis}\left(-\frac{n\pi}{3}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \Rightarrow \alpha^n + \beta^n + \gamma^n &= 4^n \operatorname{cis}\left(\frac{n\pi}{3}\right) + 4^n \operatorname{cis}\left(-\frac{n\pi}{3}\right) + \left(-\frac{1}{2}\right)^n \\
 &= 4^n \left[2 \cos\left(\frac{n\pi}{3}\right)\right] + \left(-\frac{1}{2}\right)^n \\
 &= \underline{\underline{2^{n+1} \cos\left(\frac{n\pi}{3}\right) + \left(-\frac{1}{2}\right)^n}} \checkmark
 \end{aligned}$$

# Question 4

a)  $z^5 = 1$

$$z_1 = 1$$

$$z_2 = \cos \frac{2\pi}{5} = \omega$$

$$z_3 = \cos \frac{4\pi}{5} = \omega^2$$

$$z_4 = \cos \frac{6\pi}{5} = \cos \left(-\frac{4\pi}{5}\right) = \omega^3$$

$$z_5 = \cos \frac{8\pi}{5} = \cos \left(-\frac{2\pi}{5}\right) = \omega^4$$

b) M1  $z^5 - 1 = (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)$  ①

$$= (z-1)(z^4 + z^3 + z^2 + z + 1)$$
 ②

Equating ① & ②,

$$(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4) = z^4 + z^3 + z^2 + z + 1$$

Let  $z = 1$ :

$$(1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4) = 1 + 1 + 1 + 1 + 1 = 5$$

M2

$$(1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4)$$

$$= (1-\omega-\omega^2+\omega^3)(1-\omega^3-\omega^4+\omega^7)$$

$$= 1 - \omega^3 - \omega^4 + \omega^7 - \omega + \omega^4 + \omega^5 - \omega^8 - \omega^2 + \omega^5 + \omega^6 - \omega^9 + \omega^3 - \omega^6 - \omega^7 + \omega^{10}$$

$$= 1 - \omega - \omega^2 + 2 - \omega^3 - \omega^4 + 1 \quad \text{since } \omega^5 = 1$$

$$= 4 - (\omega - \omega^2 - \omega^3 - \omega^4)$$

$$= 4 - (-1) \quad \text{since } \omega^5 = 1 \text{ as } \omega \text{ is a root of } z^5$$

$$= 5$$

$$\omega^5 - 1 = 0$$

$$(\omega-1)(1+\omega+\omega^2+\omega^3+\omega^4) = 0$$

$$1+\omega+\omega^2+\omega^3+\omega^4 = 0 \quad \text{since } \omega \neq 1$$

$$\omega + \omega^2 + \omega^3 + \omega^4 = -1$$

$$\begin{aligned}
 c) \quad (1-\omega)(1-\omega^4) &= (1-\cos \frac{2\pi}{5})(1-\cos(-\frac{2\pi}{5})) \quad \text{from (a)} \\
 &= 1 - (\cos \frac{2\pi}{5} + \cos(-\frac{2\pi}{5})) + \cos(\frac{2\pi}{5} + (-\frac{2\pi}{5})) \\
 &= 1 - 2\cos \frac{2\pi}{5} + \cos 0 \quad \text{since } \cos(-x) = \cos x \\
 &\quad \text{and } \sin(-x) = -\sin x \\
 &= 1 - 2\cos \frac{2\pi}{5} + 1 \\
 &= 2 - 2\cos \frac{2\pi}{5}
 \end{aligned}$$

$$\begin{aligned}
 d) \quad \text{Similarly, } (1-\omega^2)(1-\omega^3) &= 2 - 2\cos \frac{4\pi}{5} \\
 (1-\omega)(1-\omega^4)(1-\omega^2)(1-\omega^3) &= 5 \quad \text{from (6)} \\
 \therefore (2 - 2\cos \frac{2\pi}{5})(2 - 2\cos \frac{4\pi}{5}) &= 5 \\
 (2 - 2(1 - 2\sin^2 \frac{\pi}{5}))(2 - 2(1 - 2\sin^2 \frac{2\pi}{5})) &= 5 \quad \text{since } \cos 2x = 1 - 2\sin^2 x \\
 4\sin^2 \frac{\pi}{5} \times 4\sin^2 \frac{2\pi}{5} &= 5 \\
 (\sin \frac{\pi}{5} \sin \frac{2\pi}{5})^2 &= \frac{5}{16}
 \end{aligned}$$

$$\therefore \sin \frac{\pi}{5} \sin \frac{2\pi}{5} = \sqrt{\frac{5}{16}} = \frac{\sqrt{5}}{4}$$

$\sin \frac{\pi}{5}, \sin \frac{2\pi}{5} > 0$

$$\begin{aligned}
 e) i) M1 \quad |\vec{P_0 P_1}| &= |\omega - 1| \\
 &= |\cos \frac{2\pi}{5} - \cos 0| \\
 &= |\cos \frac{\pi}{5} (\cos \frac{\pi}{5} - \cos(-\frac{\pi}{5}))| \\
 &= |\cos \frac{\pi}{5} (2\sin \frac{\pi}{5})| \\
 &= |\cos \frac{\pi}{5}| |2\sin \frac{\pi}{5}| \\
 &= 1 \times 2 \sin \frac{\pi}{5}
 \end{aligned}$$

M2 Using the cosine rule.

$$\begin{aligned}
 |\vec{P_0 P_1}| &= \sqrt{1^2 + 1^2 - 2(1)(1)\cos \frac{2\pi}{5}} \\
 &= \sqrt{2 - 2\cos \frac{2\pi}{5}} \\
 &= \sqrt{2 - 2(1 - 2\sin^2 \frac{\pi}{5})} \quad \text{since } \cos 2x = 1 - 2\sin^2 x \\
 &= \sqrt{4\sin^2 \frac{\pi}{5}} \\
 &= 2\sin \frac{\pi}{5}
 \end{aligned}$$

ii) Similarly,  $|\vec{P_0 P_2}| = 2 \sin \left( \frac{4\pi}{5} \times \frac{1}{2} \right) = 2 \sin \frac{2\pi}{5}$

$$|\vec{P_0 P_3}| = \left| 2 \sin \left( -\frac{4\pi}{5} \times \frac{1}{2} \right) \right| = 2 \sin \frac{2\pi}{5}$$

$$|\vec{P_0 P_4}| = \left| 2 \sin \left( -\frac{2\pi}{5} \times \frac{1}{2} \right) \right| = 2 \sin \frac{\pi}{5}$$

$$|\vec{P_0 P_1}| \times |\vec{P_0 P_2}| \times |\vec{P_0 P_3}| \times |\vec{P_0 P_4}|$$

$$= \left( 2 \sin \frac{2\pi}{5} \times 2 \sin \frac{\pi}{5} \right)^2$$

$$= 16 \times \left( \frac{\sqrt{5}}{4} \right)^2$$

$$= 5$$